

5th Automotive CFD Prediction Workshop

Multi-Granularity Conformal Prediction for Reliable Neural-Operator Automotive Aerodynamic Surrogates

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For further details, please refer to our preprint: <https://arxiv.org/abs/2607.17297> Submitted to
EAAI (under review) Github:

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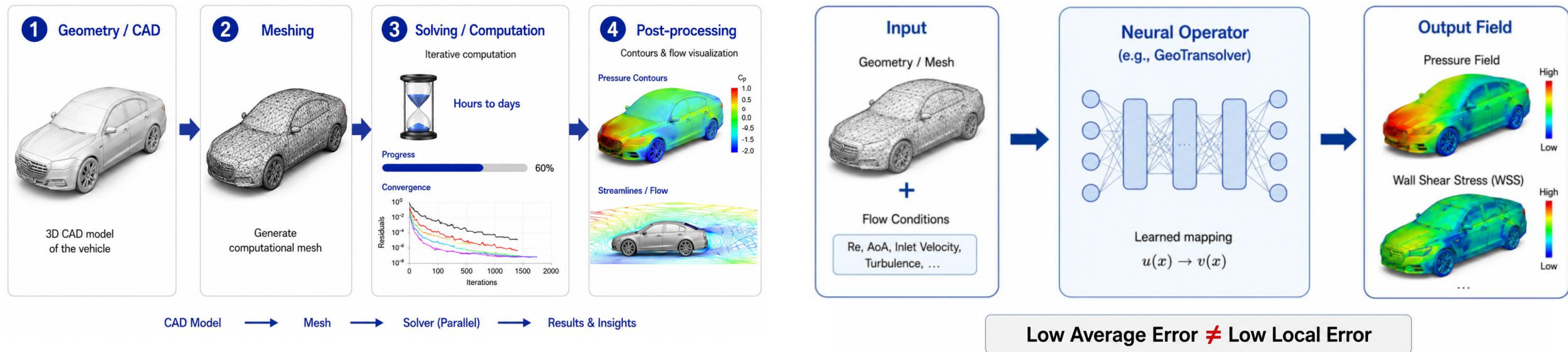
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CHAPTER 01

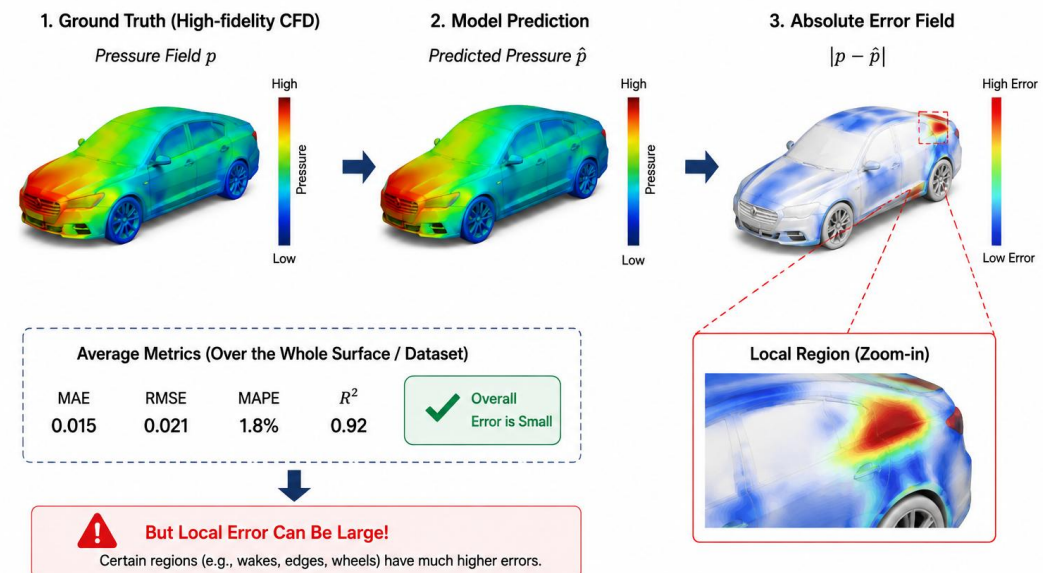
Background and motivation

Aerodynamic surrogates, uncertainty and conformal prediction

Why aerodynamic surrogates need calibrated uncertainty

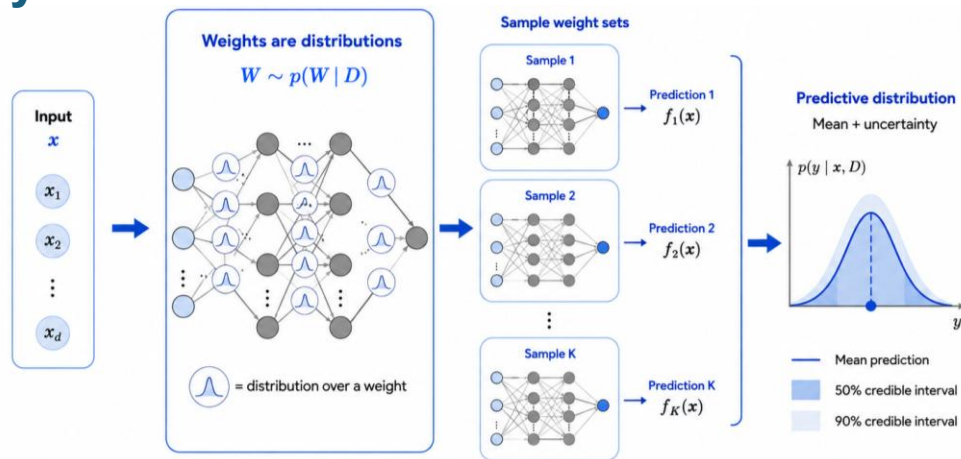


- CFD is accurate but computationally expensive.
- Neural operators enable fast aerodynamic prediction.
- Low global error may still hide large local errors, requiring calibrated uncertainty.

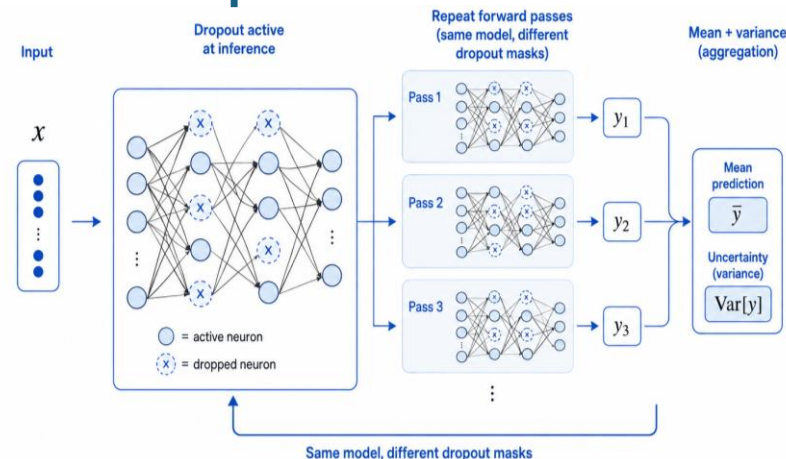


Uncertainty estimation and post-hoc calibration

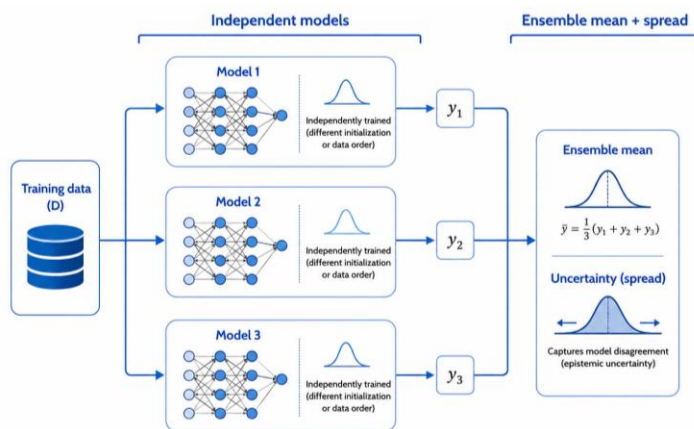
Bayesian neural networks



Monte Carlo dropout



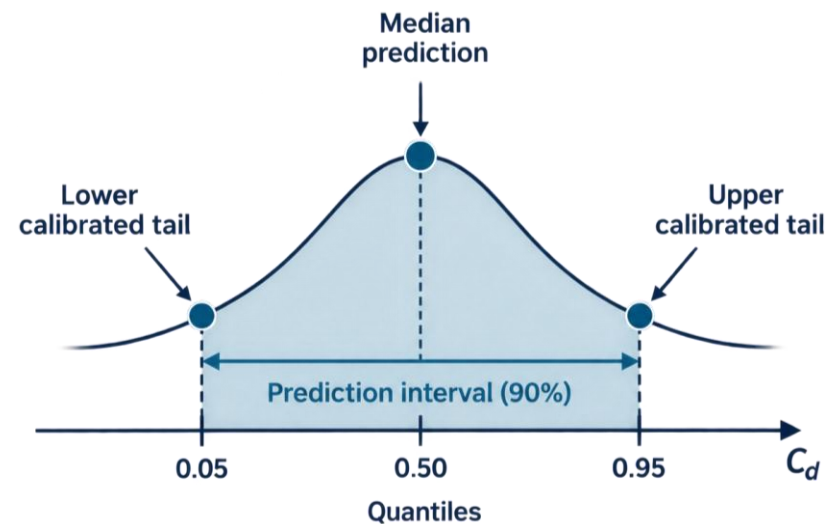
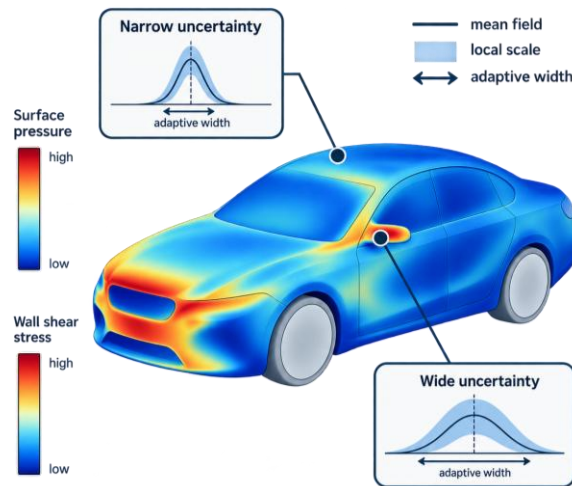
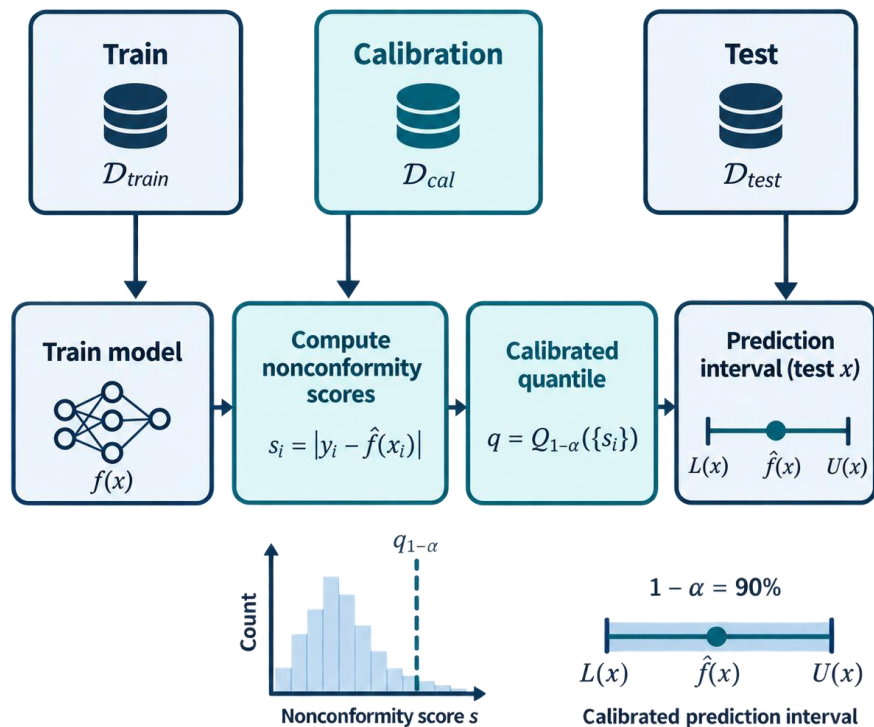
Deep ensembles



Why conformal prediction?

- BNNs require posterior approximation;
- Dropout and ensembles use repeated predictions.
- CP calibrates held-out errors and provides a coverage interpretation under exchangeability.

Conformal prediction: target and assumptions



Coverage interpretation

Standard split CP provides marginal coverage under exchangeability. It does not guarantee coverage for every vehicle or every surface point.

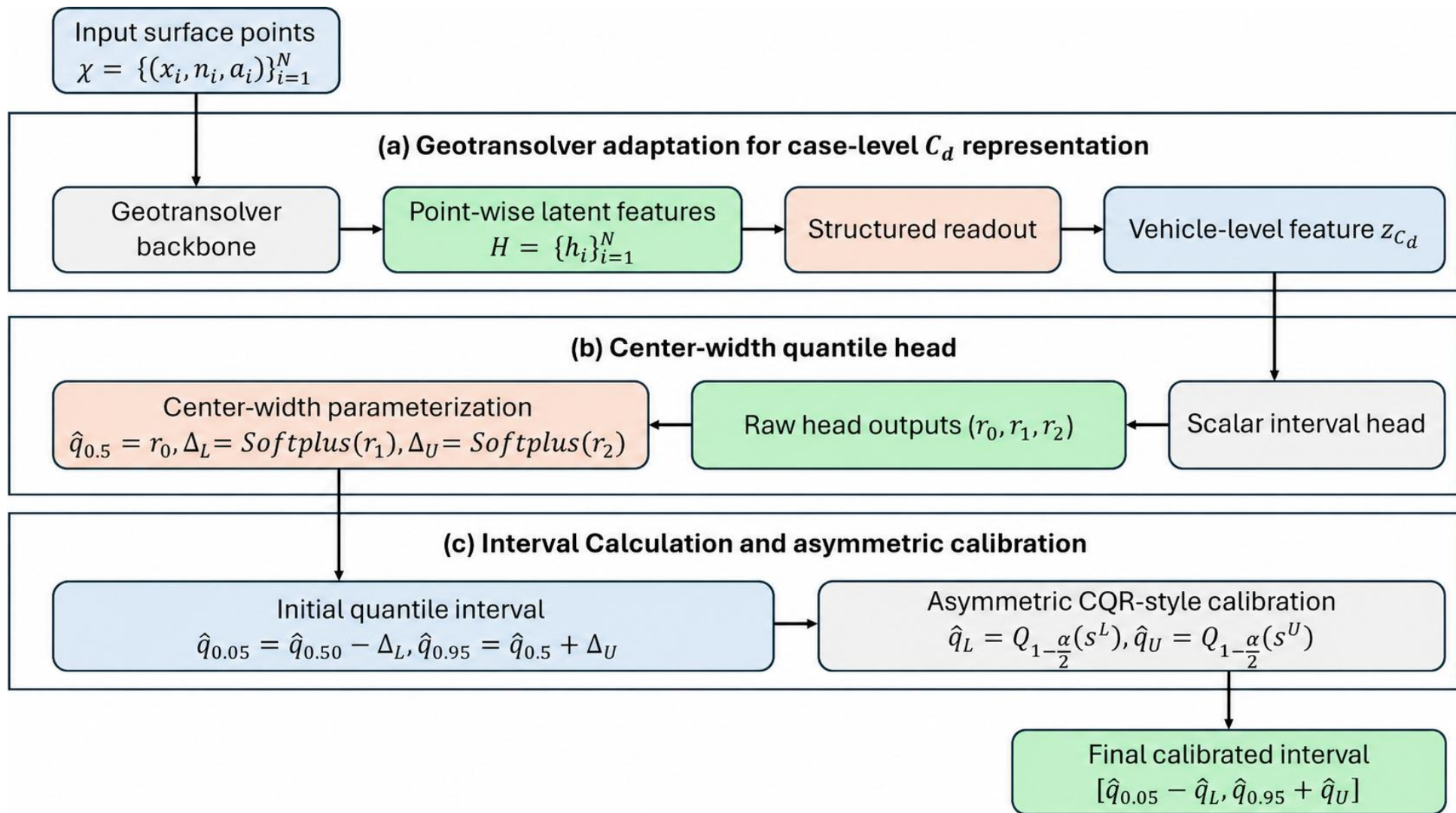
Nominal coverage: 90%.

CHAPTER 02

Conformal prediction framework

Scalar and surface calibration, residual scale and OOF aggregation

Scalar drag: structured readout and asymmetric CQR



The center-width head prevents quantile crossing. Positive-part scores preserve or expand the raw interval.

Surface fields: mean prediction and residual scale

a) Mean field prediction

Input surface representation $\mathbf{X}$,
local embedding, geometry

Geotransolver backbone

Point-wise physical field head
 f_{θ}^{mean}

Mean field prediction $\hat{\mathbf{y}} \in \mathbb{R}^{N \times 4}$

Channels $[p, \tau_x, \tau_y, \tau_z]$

b) Residual scale head

Sigma-head input

Local geometric features $[\mathbf{x}_i, \mathbf{n}_i]$

Detached mean prediction $sg(\hat{\mathbf{y}}_i)$

Concatenate $[\mathbf{x}_i, \mathbf{n}_i, sg(\hat{\mathbf{y}}_i)]$

Sigma MLP Linear + GELU $\rightarrow$ Linear

Positive scale output
 $\hat{\sigma} = \text{Softplus}(\text{raw}) + \sigma_{min}$

c) Training and calibration

$$\tilde{r} = \max(|y - sg(\hat{\mathbf{y}})|, r_{min})$$

Training losses

$$\mathcal{L}_{mean} = \text{MSE}(\hat{\mathbf{y}}, \mathbf{y})$$

$$\mathcal{L}_{scale} = \text{SmoothL1}(\log \hat{\sigma}, \log \tilde{r})$$

$$\mathcal{L}_{smooth} = \text{KNN smoothness on } \log(\hat{\sigma})$$

Residual-normalized score

$$s = \frac{|\mathbf{y} - \hat{\mathbf{y}}|}{\hat{\sigma} + \epsilon}$$

Global absolute

Point-adaptive
normalized

Case-wise normalized

Final interval(no scale)

Final interval(scaled)

Final interval(scaled)

Three calibration granularities for surface fields

Method	Calibration score / aggregation	Interval half-width
Global absolute	Pool $ y_{b,i,c} - \hat{y}_{b,i,c} $ within each channel	$\hat{q}_c^{global}$ (constant)
Point-adaptive	Pool $\frac{ y_{b,i,c} - \hat{y}_{b,i,c} }{\hat{\sigma}_{b,i,c} + \varepsilon}$	$\hat{q}_c^{point} \hat{\sigma}_{i,c}$ (spatial)
Case-wise	90th percentile within each case; then 90th percentile across cases	$\hat{q}_c^{case} \hat{\sigma}_{i,c}$ (spatial)

The learned scale controls spatial variation; the calibration multiplier controls conservativeness.

Case-wise aggregation uses a within-case quantile, not a maximum or a simultaneous all-points guarantee.

Global absolute calibration

$$a_{b,i,c} = |y_{b,i,c} - \hat{y}_{b,i,c}| \quad (1)$$

$$\hat{q}_c^{global} = Q_{1-\alpha} \left(\{a_{b,i,c}\}_{b,i} \right) \quad (2)$$

$$\mathcal{J}_{i,c}^{global} = \left[\hat{y}_{i,c} - \hat{q}_c^{global}, \quad \hat{y}_{i,c} + \hat{q}_c^{global} \right] \quad (3)$$

Point-adaptive normalized calibration

$$\hat{q}_c^{point} = Q_{1-\alpha} \left(\{s_{b,i,c}^{field}\}_{b,i} \right) \quad (4)$$

$$\mathcal{J}_{i,c}^{point} = \left[\hat{y}_{i,c} - \hat{q}_c^{point} \hat{\sigma}_{i,c}, \quad \hat{y}_{i,c} + \hat{q}_c^{point} \hat{\sigma}_{i,c} \right] \quad (5)$$

Case-wise normalized calibration

$$s_{b,c}^{case} = A_{case} \left(\{s_{b,i,c}^{field}\}_{i=1}^N \right) \quad (6)$$

$$s_{b,c}^{case} = Q_{1-\alpha} \left(\{s_{b,i,c}^{field}\}_{i=1}^N \right) \quad (7)$$

$$\hat{q}_c^{case} = Q_{1-\alpha} \left(\{s_{b,c}^{case}\}_b \right) \quad (8)$$

$$\mathcal{J}_{i,c}^{case} = \left[\hat{y}_{i,c} - \hat{q}_c^{case} \hat{\sigma}_{i,c}, \quad \hat{y}_{i,c} + \hat{q}_c^{case} \hat{\sigma}_{i,c} \right] \quad (9)$$

Why CV-assisted OOF calibration?

Limited CFD labels

Only 400 training–calibration cases. A single split holds out 80 cases for Cd or 100 for surface fields.

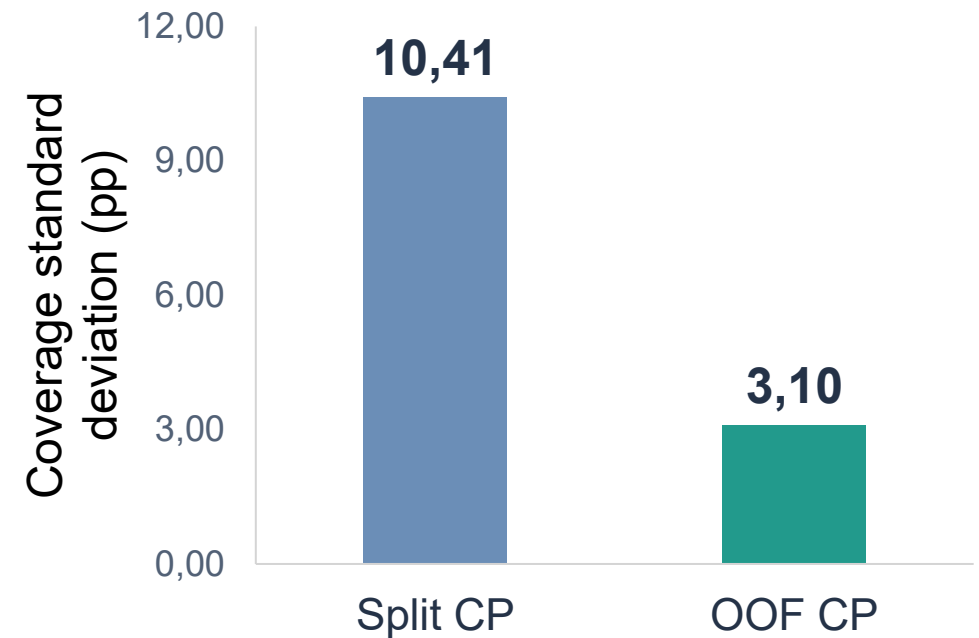
Less dependence on one split

Each case contributes a score from a model that excluded it. Pooling scores across folds reduces calibration sensitivity to the chosen partition.

A final predictor trained on all 400 cases

After collecting OOF scores, refit on the full pool. Better point accuracy comes from this refit, while score pooling stabilizes calibration.

Evidence: scalar Cd resampling



500 calibration–evaluation resamples

Trade-off: mean width increases from 25.00 to 31.60 ($\times 10^{-3}$).

CHAPTER 03

Experimental setup

Dataset, training configuration and evaluation metrics

DrivAerML dataset and calibration protocols

Protocol	Training / calibration	Final evaluation
Official split	400-case training–calibration pool	34 validation; 50 test
Scalar OOF	5 folds: 320 train / 80 held out	Refit on all 400 cases
Field OOF	4 folds: 300 train / 100 held out	Refit on all 400 cases
Smoothness ablation	Separate random 80/10/10 split	49-case test subset

Validation supports model selection only; it is excluded from calibration and final testing.

Scalar stability: 500 disjoint 200/200 calibration–evaluation resamples using cached predictions, without retraining.

Evaluation: interval efficiency and vehicle reliability

Scalar drag coefficient

$$\text{Coverage} = \frac{1}{n_{\text{test}}} \sum_{j=1}^{n_{\text{test}}} \mathbb{I}\{y_j \in [L_j, U_j]\} \quad (10)$$

$$\text{Width} = \frac{1}{n_{\text{test}}} \sum_{j=1}^{n_{\text{test}}} (U_j - L_j) \quad (11)$$

$$\text{IS}_{\alpha,j} = (U_j - L_j) + \frac{2}{\alpha} (L_j - y_j) \mathbb{I}\{y_j < L_j\} + \frac{2}{\alpha} (y_j - U_j) \mathbb{I}\{y_j > U_j\} \quad (12)$$

Surface fields

$$c \in \{p, \tau_x, \tau_y, \tau_z\}$$

$$\text{Coverage}_c = \frac{1}{BN} \sum_{b=1}^B \sum_{i=1}^N \mathbb{I}\{y_{b,i,c} \in [L_{b,i,c}, U_{b,i,c}]\} \quad (13)$$

$$\text{Width}_c = \frac{1}{BN} \sum_{b=1}^B \sum_{i=1}^N (U_{b,i,c} - L_{b,i,c}) \quad (14)$$

B: test vehicles; N: inference points per vehicle. Report interval score averaged over test samples.

Also report vehicles below 90% coverage and P5 of case-level coverage. Surface points are spatially correlated.

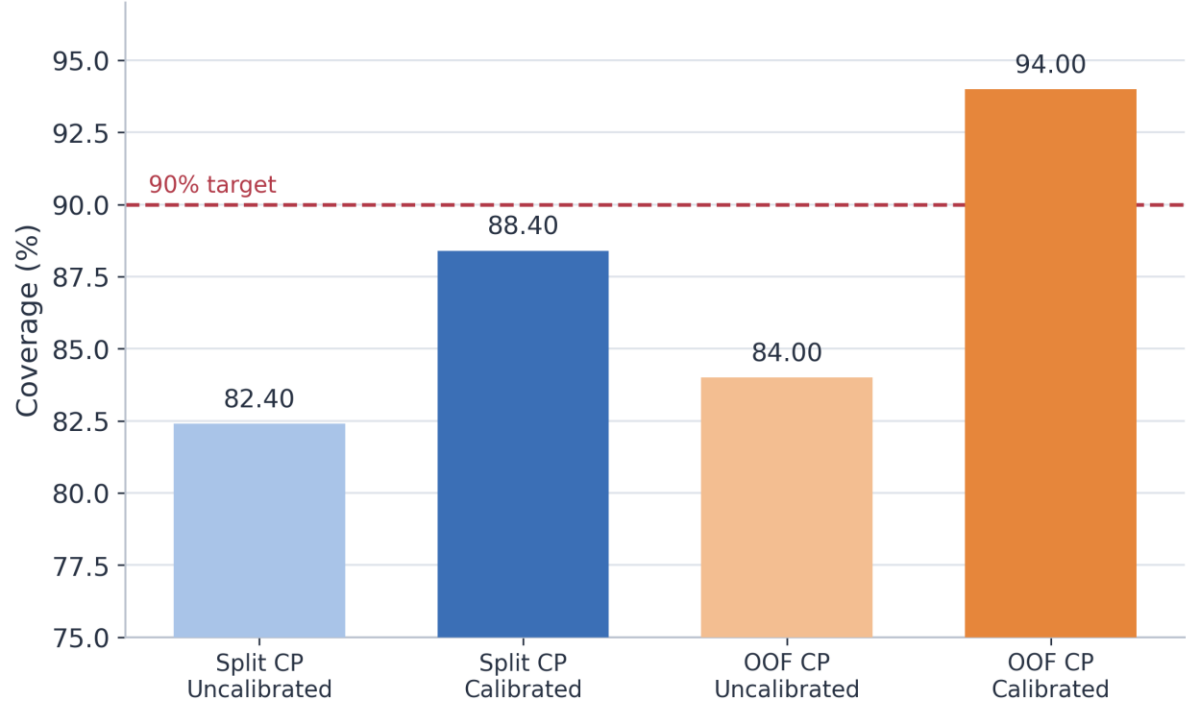
CHAPTER 04

Drag Coefficient Results

Scalar reliability, smoothness and calibration granularity

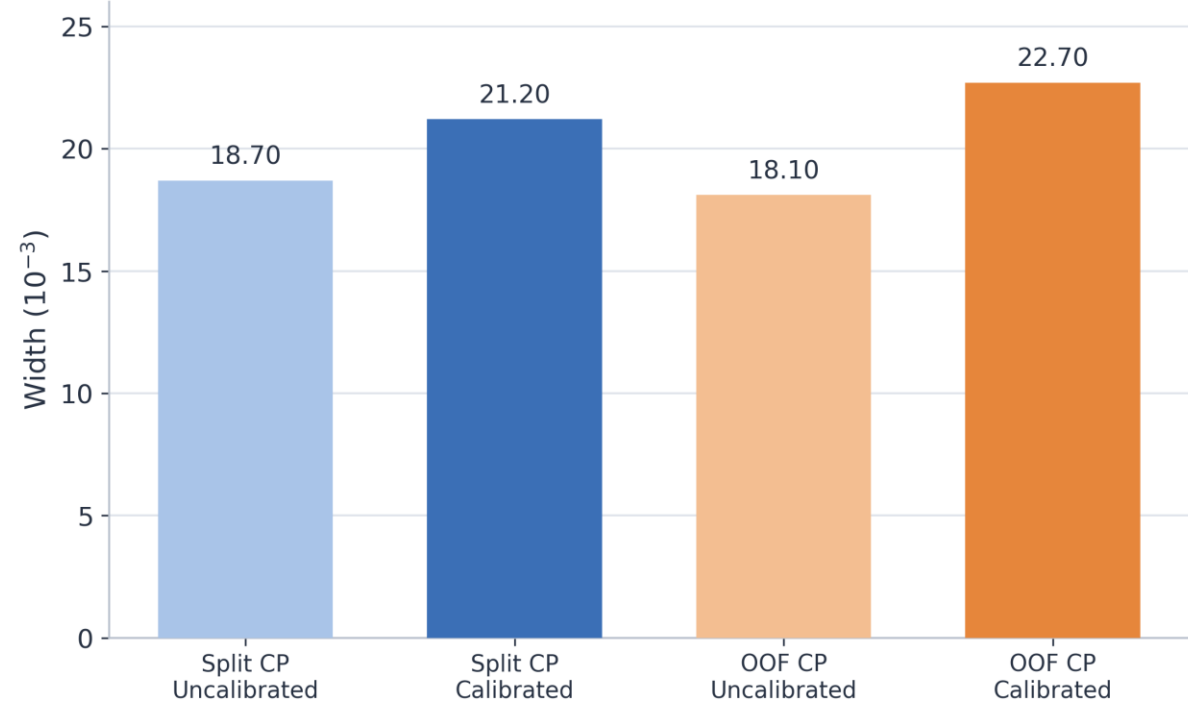
Scalar calibration: test coverage and interval width

Prediction-Interval Coverage



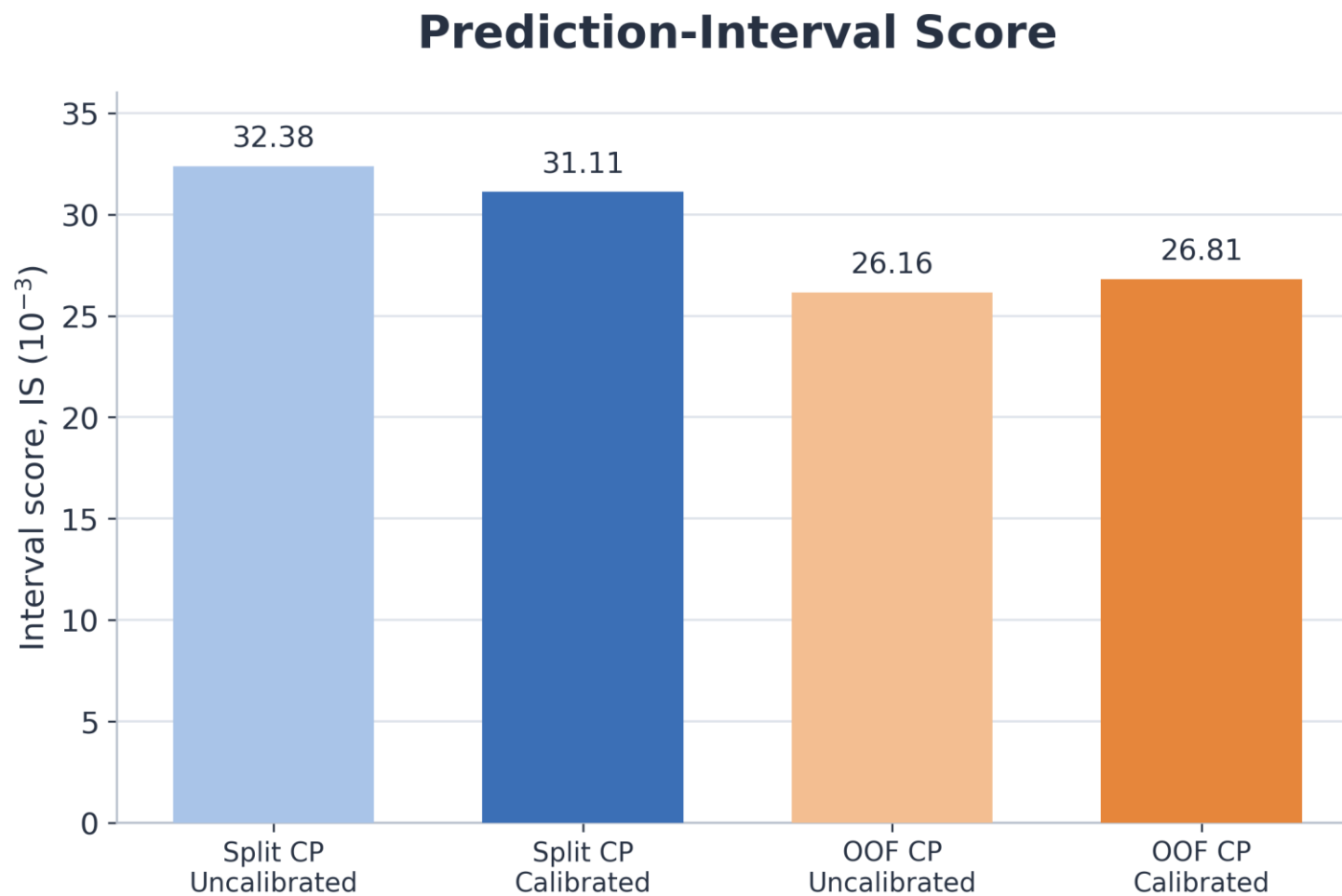
Calibrated OOF CP is the only setting above the nominal 90% target.

Mean Prediction-Interval Width



Narrower intervals indicate greater sharpness.

Scalar calibration: interval score



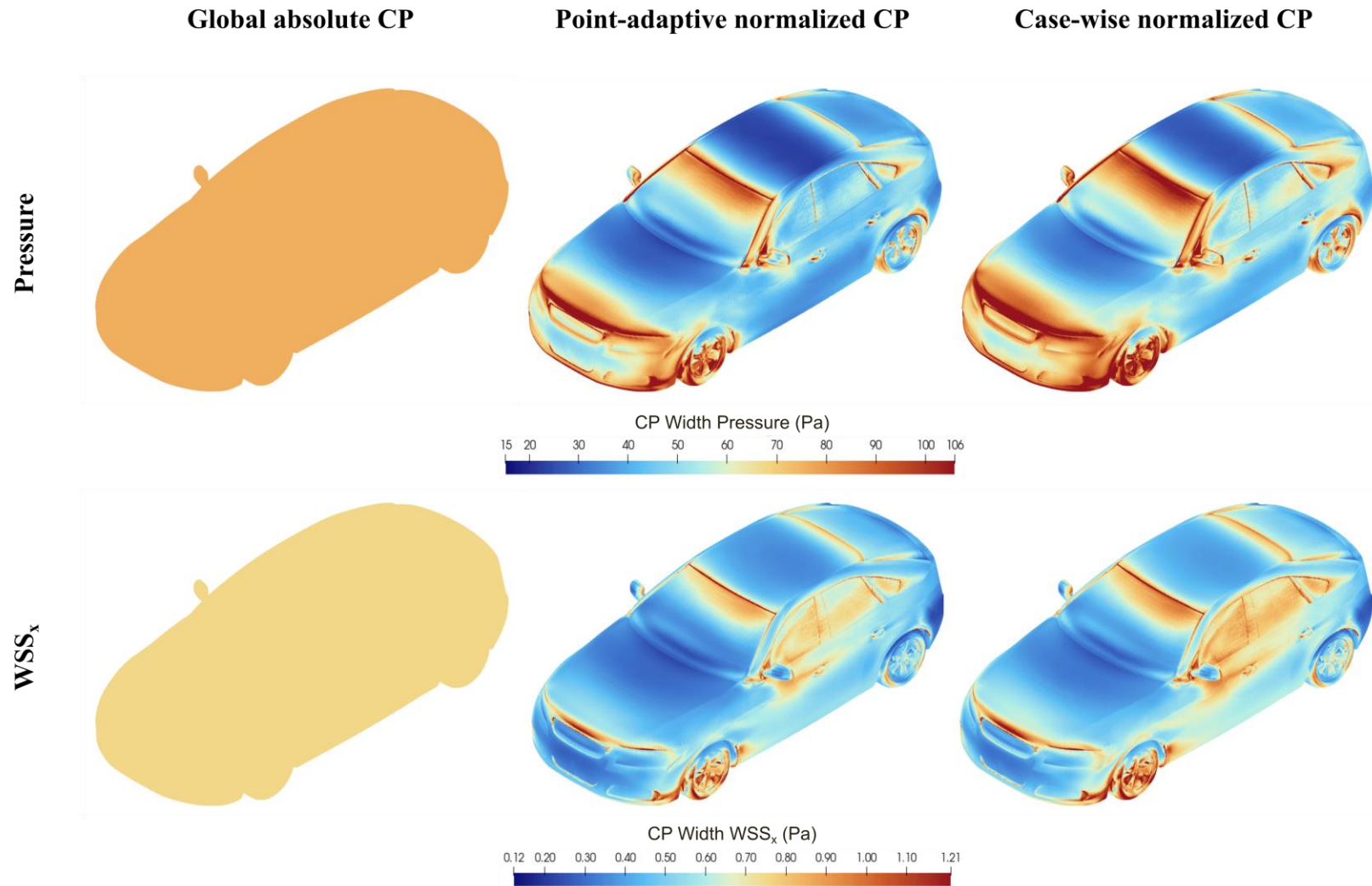
Lower interval scores indicate a better balance between coverage and sharpness.

CHAPTER 05

Surface Field Results

Vehicle-level coverage, spatial intervals and Transolver

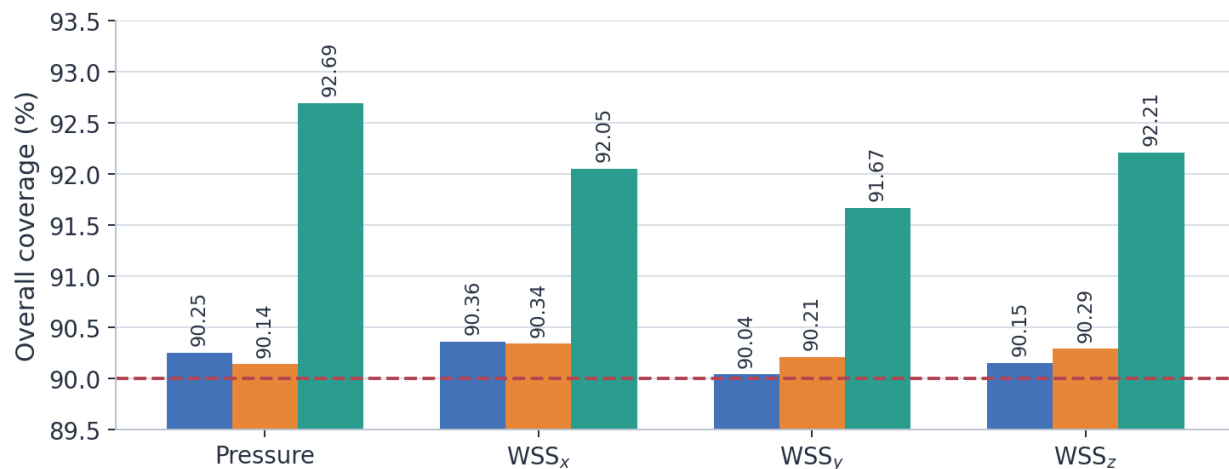
Spatial interval widths identify uncertain surface regions



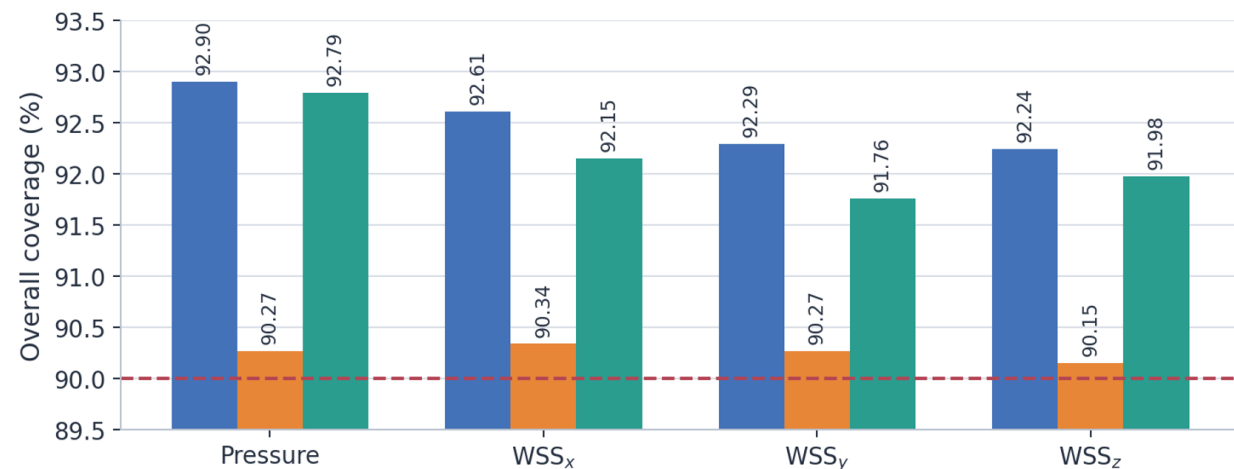
Calibration granularity: Comparison

Conformal Coverage

Global absolute CP Case-wise normalized CP
Point-adaptive normalized CP 90% target



Ordinary Split CP



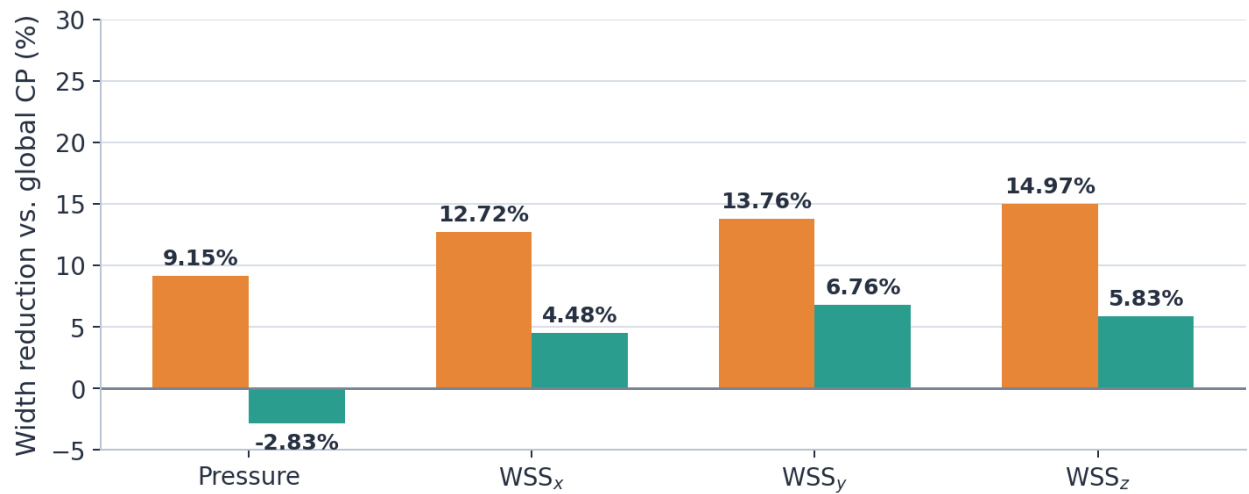
CV-assisted OOF

Case-wise normalized CP is conservative; point-adaptive CP remains closest to the 90% target.

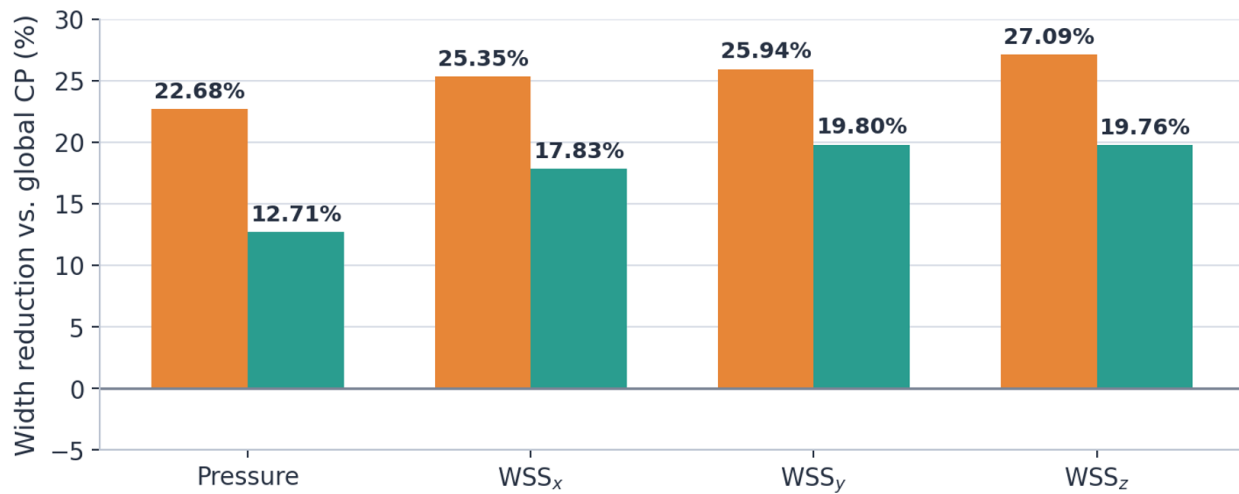
Interval-width reduction: Comparison

Interval-Width Reduction

Point-adaptive normalized CP Case-wise normalized CP



Ordinary Split CP



CV-assissted OOF

Point-adaptive CP reduces width by 9.15%–14.97%; case-wise CP is more conservative.

Vehicle-level coverage reveals the efficiency trade-off

OOF: cases below 90%	Pressure	WSS_x	WSS_y	WSS_z
Global absolute	2 / 50	2 / 50	1 / 50	1 / 50
Point-adaptive	15 / 50	15 / 50	17 / 50	19 / 50
Case-wise	6 / 50	4 / 50	4 / 50	4 / 50

Case-wise vs. point-adaptive

Case-wise calibration reduces low-coverage vehicles while retaining spatially varying intervals.

Global vs. normalized

Global CP has the fewest low-coverage vehicles under OOF, but produces wider, spatially uniform intervals.

CHAPTER 06

Engineering implications and conclusions

CFD decision support, limitations and main findings

Conclusions and engineering implications

Scalar reliability and screening

- Asymmetric CQR corrects under-coverage.
- OOF improves stability but widens intervals.
- Use wide or target-overlapping intervals to trigger CFD verification.

Calibration choice and transfer

- Point-adaptive CP gives the narrowest intervals.
- Case-wise CP improves vehicle-level reliability.
- Calibration behavior transfers across backbones.

Spatial uncertainty

- Residual normalization enables spatially adaptive widths.
- Smoothness reduces local scale variation by 74.29%.
- Large widths highlight less reliable surface regions.

Engineering use and limits

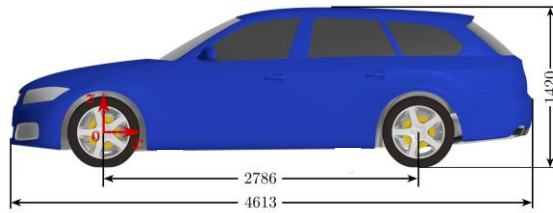
- Use calibrated intervals to prioritize CFD verification.
- Coverage relies on exchangeability.
- OOF gains also reflect full-data refitting.

DrivAerRim

A full-vehicle dataset with rotating wheels for aerodynamic design of wheel rims

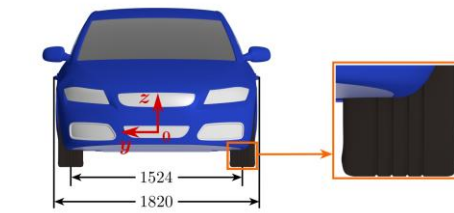
<https://github.com/Beyond0619/DrivAerRim>

<https://huggingface.co/datasets/BeyondXia1212/DrivAerRim/tree/main>

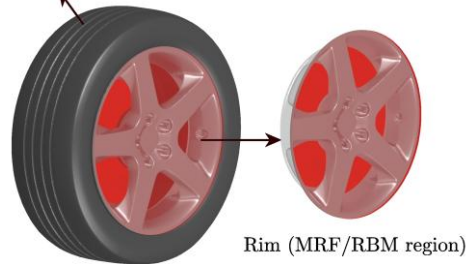


□ DrivAer Estateback

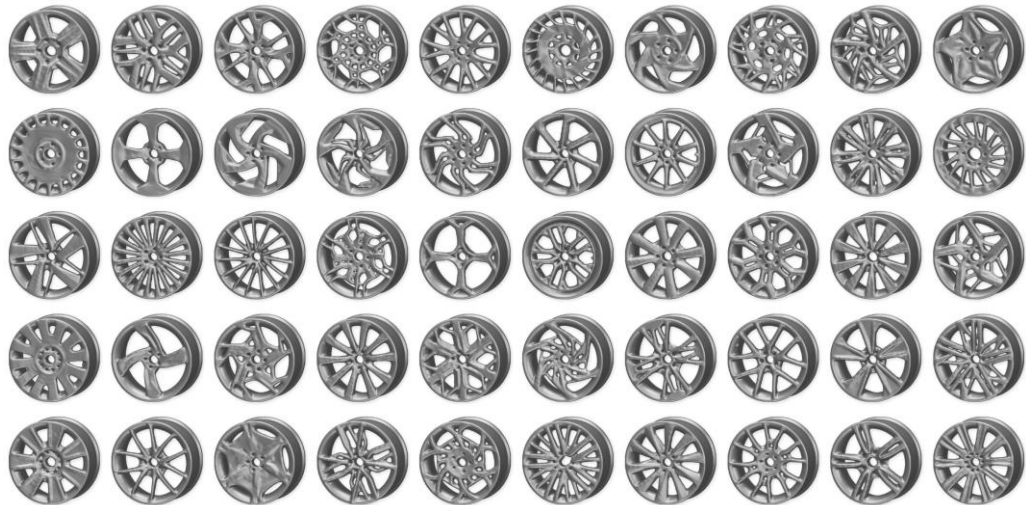
- ◆ DrivAerRim:
RANS+MRF (Open-sourced)
- ◆ DrivAerRim++:
IDDES+RBM (To be released)



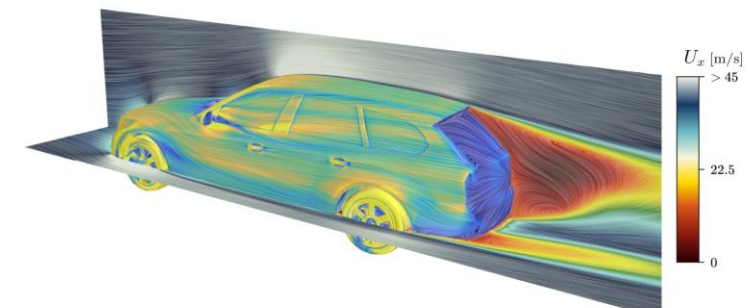
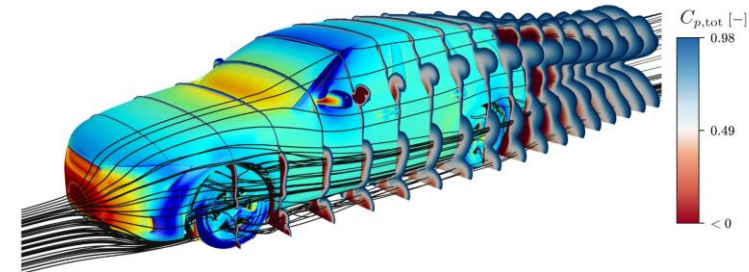
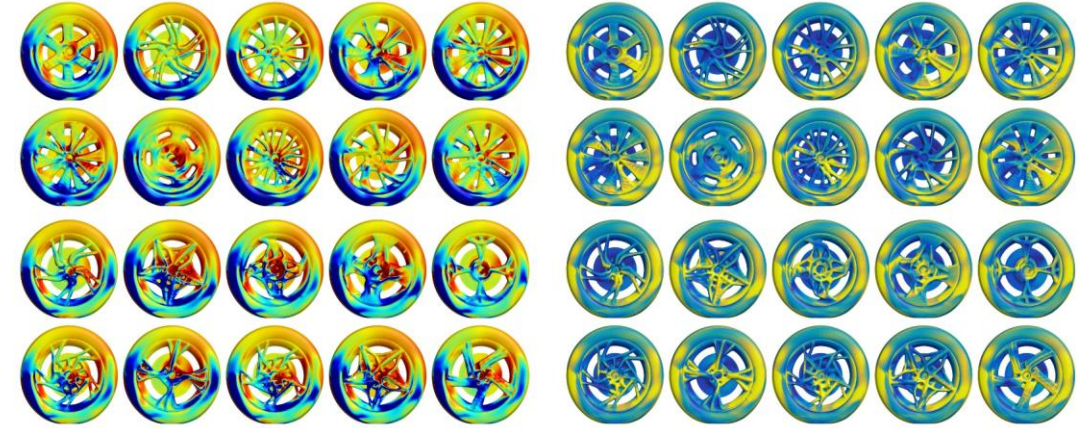
Trye (Rotating Wall)



Rim (MRF/RBM region)



□ Over 900 Rim configurations



Thanks for listening!

Acknowledgement

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Q & A

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